Kabadi Connect: supporting document for Bhashini API key request

Live demo: https://d7oz11dsxwlh1.cloudfront.net | Method page: https://d7oz11dsxwlh1.cloudfront.net/methodology

Project

Kabadi Connect is a hackathon project (WeMakeDevs x AWS Environmental Hacks, Waste and Energy track) built by a student at Thapar Institute of Engineering and Technology, Patiala. Households post a pickup request for recyclable waste (paper, plastic, metal, e-waste, glass). Requests are batched by area, routed, and handed to a local kabadiwala (informal scrap collector), who records the real weights. A dashboard shows kg kept out of landfill, CO2e avoided and segregation quality.

Why voice

Many kabadiwalas read little and use low-cost phones. v2 replaces email and password login with a printed QR card, and adds a Hindi voice assistant: the collector holds a button and says 'agla ghar kahan hai', 'paanch kilo kagaz' or 'ghar par koi nahi'. The app answers in Hindi and asks for a spoken or tapped yes before saving anything.

How Bhashini would be used

ASR converts the collector's short Hindi clip (usually under 10 seconds) to text. TTS reads the reply back. Calls are made from our server (AWS Lambda), never from the browser. The API key is stored encrypted (AWS SSM Parameter Store, SecureString). Audio is not stored after transcription. Expected volume: about 200 calls per day during the hackathon and a small pilot. If Bhashini is slow or unavailable, the app falls back to AWS speech services, so Bhashini is the primary provider and the quality is monitored per call.

Languages

Hindi now (Punjabi also requested). More Indian languages are planned through Bhashini so collectors in other regions can use the app in their own language.

Privacy

Voice clips and photos are processed in memory and not stored. Monthly statements and impact totals contain no household addresses or phone numbers.

Architecture (summary)

React PWA on S3 and CloudFront; Amazon API Gateway with Cognito; AWS Lambda; DynamoDB; SQS and EventBridge for dispatch; Amazon Location Service for route ordering; Amazon Bedrock for intent understanding; Amazon Polly and Transcribe as speech fallback.